

Management of Gout

Acute flare → treat-and-tap · chronic → treat-to-target urate <6. ACR 2020 guideline.

Diagnosis & Mimics

- Acute flare: abrupt warm, red, exquisitely painful joint ± fever; ~73% at the 1st MTP.
- **Tap before steroids** if infection possible — crystals do NOT exclude septic arthritis (they can coexist).

Synovial WBC	Interpretation
<2K	Non-inflammatory
2–75K	Inflammatory (gout; MSU crystals)
>75K	High risk for septic (overlap: gout can >100K; septic lower post-abx)

- Gout = needle, **negatively** birefringent MSU. Pseudogout = rhomboid, weakly positive CPP (knee >50%).
- Hemarthrosis = sanguineous fluid.

Acute Flare — Pick by Comorbidity

Agent	Dose	Avoid / caution
NSAID	Naproxen 500 BID; indometh 50 TID ×5–7d (all equal)	GI bleed, CKD, ischemic CAD
Steroid	Pred 40 mg ×5d; IA triamcinolone 40/20/10 by joint	DM (glucose), active infection
Colchicine	1.2 → 0.6 mg in 1h → 0.6 QD–BID	Only <48h; renal dose; statin/CYP3A4; diarrhea
Anakinra	100 mg SQ daily ×3–5d (IL-1 RA)	Active infection; w/ rheum if others c/i

Modifiable Risk Factors

- **Diet ↑**: red meat, shellfish, beer/spirits, fructose. **Protective**: dairy.
- **Meds ↑**: thiazide/loop diuretics, ASA, ACE-I, ARBs (except **losartan**), ethambutol, PZA, niacin.
- **Prefer** losartan + amlodipine for BP (mild uricosuric). Treating HTN/DM/HLD lowers risk.
- **Do NOT** stop indicated low-dose aspirin to manage gout (ACR 2020).

Urate-Lowering Therapy (ULT)

Start ULT if	Also consider after 1 flare if
≥2 flares/yr · tophi · 1 flare + erosions	CKD ≥3 · urate >9 · urolithiasis

- **Allopurinol first-line** (all, incl. CKD): start 100 mg/d (≤50 if CKD), titrate q2–4 wk, max ~800.
- **Treat-to-target urate <6 mg/dL** (<5 to dissolve tophi).
- **Prophylaxis** (low-dose colchicine/NSAID/prednisone) for 3–6 mo until target dose reached.
- Starting ULT during a flare does NOT worsen it; continue chronic ULT through flares.
- **Febuxostat 2nd-line only** — CARES boxed warning (CV/all-cause death).

HLA-B*5801 → allopurinol hypersensitivity (rash, fever, eosinophilia, ↑LFTs, renal failure). Screen Han Chinese / Korean / Thai / African-American ancestry; if positive, do NOT use allopurinol.

Do / AVOID

- **DO** arthrocentesis + X-ray for any new hot joint; blood cultures if febrile.
- **DO** match the flare drug to comorbidities; pair every ULT start with prophylaxis.
- **AVOID** trusting serum urate during a flare — may be falsely low, normal, or high.
- **AVOID** steroids during active infection; treat gout after source control.
- **AVOID** stopping ULT during a flare.

Pearl: recent skin infection + a hot joint with a high cell count = treat for septic arthritis even when MSU crystals are present.